



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

AI Agents Fundamentals for Beginners

Course Code: C690

Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	5 October 2026	Non-WSQ edition of AI Agents Fundamentals for Beginners (C690): 212 slides, LG, LP and 12 labs; 9:30am - 5:30pm schedule, 7.5 instructional hours per day.	Tertiary Infotech Academy Pte Ltd

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Course identity and delivery allocation

AI Agents Fundamentals for Beginners | C690 | v1.0. Two days, 15 instructional hours: 450 minutes per day, 9:30am - 5:30pm, with a single 30-minute lunch break. Each session below was rescaled proportionally from the source plan so the topics keep their relative weight.

Day	Instruction	Lunch break	Day schedule
1	450 minutes	30 minutes	9:30am - 5:30pm
2	450 minutes	30 minutes	9:30am - 5:30pm
Total	900 minutes (15 hours)	60 minutes	2 days

Day 1 • 9:30am - 5:30pm • slides 1 to 106

Session	Minutes	Slides	Method / evidence
Topic 1 teaching + administration	115	8-56	Mechanism explanation, worked artifacts, comparison and discussion
Lab 1: Workflow business case	30	48-50	Guided practice, evidence capture and peer review
Lab 2: Tokenize business tickets	40	51-53	Guided practice, evidence capture and peer review
Lab 3: Embeddings and similarity	40	54-56	Guided practice, evidence capture and peer review
Topic 2 teaching + administration	110	57-105	Mechanism explanation, worked artifacts, comparison and discussion
Lab 4: Grounded policy retrieval	35	97-99	Guided practice, evidence capture and peer review
Lab 5: Bounded tool agent	40	100-102	Guided practice, evidence capture and peer review
Lab 6: Support approval workflow	40	103-105	Guided practice, evidence capture and peer review

The day runs 9:30am - 5:30pm: 450 instructional minutes plus one 30-minute lunch break. The trainer allocates the teaching block across the topic slides and the opening and closing pages; depth is prioritised through worked examples rather than reading every slide aloud.

Day 2 • 9:30am - 5:30pm • slides 107 to 212

Session	Minutes	Slides	Method / evidence
Topic 3 teaching + administration	100	113-161	Mechanism explanation, worked artifacts, comparison and discussion
Lab 7: Sales evidence and consent	35	153-155	Guided practice, evidence capture and peer review
Lab 8: Invoice exception agent	40	156-158	Guided practice, evidence capture and peer review
Lab 9: HR and multi-agent handoffs	35	159-161	Guided practice, evidence capture and peer review
Topic 4 teaching + administration	100	162-210	Mechanism explanation, worked artifacts, comparison and discussion
Lab 10: Train a support classifier	70	202-204	Guided practice, evidence capture and peer review
Lab 11: Agent evaluation scorecard	35	205-207	Guided practice, evidence capture and peer review
Lab 12: Governance and incident recovery	35	208-210	Guided practice, evidence capture and peer review

The day runs 9:30am - 5:30pm: 450 instructional minutes plus one 30-minute lunch break. The trainer allocates the teaching block across the topic slides and the opening and closing pages; depth is prioritised through worked examples rather than reading every slide aloud.

Lab facilitation and acceptance

Lab 1: Workflow business case - slides 48-50

Select a bounded business workflow and calculate net capacity released.

Alignment: LO1

Evidence: Support baseline releases 83.33 hours before additional overhead. The proposed pilot includes an owner and an explicit action boundary. A sensitivity result and quality floor accompany the financial claim.

Resources: labs/lab-01-workflow-business-case; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 2: Tokenize business tickets - slides 51-53

Use a real Hugging Face tokenizer and explain IDs, masks, subwords and unknown terms.

Alignment: LO2

Evidence: Real Hugging Face BertTokenizerFast output exists. Token IDs and masks have equal length for every example. The learner explains unknown tokens, subword boundaries and truncation.

Resources: labs/lab-02-tokenize-business-tickets; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 3: Embeddings and similarity - slides 54-56

Represent tokens with a transformer and compare cosine with Euclidean similarity.

Alignment: LO2

Evidence: The real encoder output shape is recorded. $\text{Cosine}(q,d1)=1$ and $\text{cosine}(q,d2)$ is about 0.7071. The report distinguishes encoder representation from autoregressive text generation.

Resources: labs/lab-03-embeddings-and-similarity; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 4: Grounded policy retrieval - slides 97-99

Retrieve supporting policy clauses and abstain when evidence is absent.

Alignment: LO2

Evidence: Every asserted rule has a supporting policy ID. The missing shipping guarantee triggers abstention. Injected document instructions do not grant authority.

Resources: labs/lab-04-grounded-policy-retrieval; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 5: Bounded tool agent - slides 100-102

Run a model-directed tool loop with schema checks and a strict read-only boundary.

Alignment: LO2

Evidence: Allowed lookup returns O-104 with a source record. Cross-customer lookup and write tools are rejected. The final answer relies on actual tool results and stops within six rounds.

Resources: labs/lab-05-bounded-tool-agent; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 6: Support approval workflow - slides 103-105

Prepare a refund response and demonstrate approval-bound state transitions.

Alignment: LO4

Evidence: Missing receipt does not become an approved refund. The repeated request does not create a second ledger operation. Approval is tied to the exact action payload.

Resources: labs/lab-06-support-approval-workflow; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 7: Sales evidence and consent - slides 153-155

Produce sourced lead briefs and consent-respecting outreach drafts.

Alignment: LO4

Evidence: L-01 score is 75; L-02 score is 40. No-contact records are marked `research_only`. Every marketing claim is from the approved claim file.

Resources: labs/lab-07-sales-evidence-and-consent; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 8: Invoice exception agent - slides 156-158

Perform deterministic three-way matching and explain exceptions with evidence.

Alignment: LO4

Evidence: INV-100 variance is 50.00. Duplicate and missing-PO records are flagged. The memo does not authorise payment.

Resources: labs/lab-08-invoice-exception-agent; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 9: HR and multi-agent handoffs - slides 159-161

Coordinate two bounded specialist roles and reconcile their outputs.

Alignment: LO4

Evidence: Operations staff do not receive manager-only approvals. The handoff preserves `task_id` and source references. The final checklist resolves or escalates conflicting evidence.

Resources: labs/lab-09-hr-and-multi-agent-handoffs; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 10: Train a support classifier - slides 202-204

Train and compare real Hugging Face classifiers and a transparent baseline.

Alignment: LO3

Evidence: Training runs use real gradients and save a reloadable model. Training/validation/test IDs and texts are disjoint. Two strategies and a count baseline are compared; selection uses validation only.

Resources: labs/lab-10-train-a-support-classifier; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 11: Agent evaluation scorecard - slides 205-207

Compute task-quality and cost metrics with explicit denominators.

Alignment: LO3

Evidence: Precision=0.75, recall=0.90, accuracy=0.92. Cost per success includes the cost of failed runs. Release recommendation includes failure evidence, not only averages.

Resources: labs/lab-11-agent-evaluation-scorecard; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 12: Governance and incident recovery - slides 208-210

Test denied actions, approval binding and a stop/recovery plan.

Alignment: LO4

Evidence: Unapproved writes and cross-customer reads are denied by code. Stop state prevents execution regardless of model output. Recovery checks transaction state before retry and assigns an owner.

Resources: labs/lab-12-governance-and-incident-recovery; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Preparation and release controls

Verify package version, install the three model-lab environments and execute all verification scripts before delivery. Record technical failures and provide an equivalent controlled rerun rather than fabricating evidence.